

TYPE THEORY AS A TOPIC OF MATHEMATICAL STUDY

AN INTRODUCTION TO GENERALISED ALGEBRAIC THEORIES

Formal proof and synthetic mathematics workshop – June 26th 2026

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Or, to be precise, worked *in* type theories.

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But:

- What is a type theory?
- What is a model of such a type theory?
- What does it mean “to work in” such a type theory?

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- What does it mean “to work in” such a type theory?

These questions can be given *mathematical* answers

WARMUP: UNIVERSAL ALGEBRA

Who here has been exposed to **universal algebra**?

Ambient logic = set-theoretic notation (but could be a type theory!): $\in$, Set , $\prod$, $\sum$...

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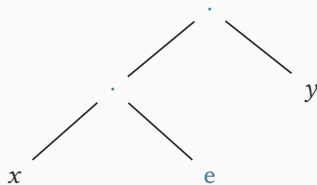
Signature

A signature Σ consists of:

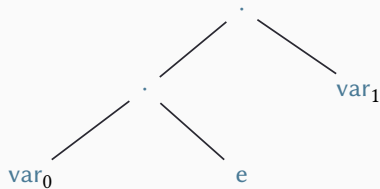
- a set of operations (generators) op_Σ , specified by an arity $\text{ar}_\Sigma \in \text{op}_\Sigma \rightarrow \mathbb{N}$

Signature of monoids :

- $\text{op} := \{e, \cdot\}$, $\text{ar}(e) := 0$, $\text{ar}(\cdot) := 2$



- They exist!
- Nice representation in (functional) programming languages (algebraic datatypes)



- They exist!
- Nice representation in (functional) programming languages (algebraic datatypes)
- Variables complicate things, but there are ways

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Signature

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- a set of operations (generators) op_Σ , specified by an arity $ar_\Sigma \in op_\Sigma \rightarrow \mathbb{N}$
→ generates a set of terms $Tm \in \mathbb{N} \rightarrow \mathbf{Set}$

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→ generates a set of terms $\text{Tm} \in \mathbb{N} \rightarrow \text{Set}$
- for each n , a set of equations (relations) $\text{eqs}_\Sigma(n) \subseteq \text{Tm}(n) \times \text{Tm}(n)$

Signature of monoids :

- $\text{op} := \{e, \cdot\}$, $\text{ar}(e) := 0$, $\text{ar}(\cdot) := 2$
- $e \in \text{Tm}(0)$, $(x \cdot e) \cdot y \in \text{Tm}(2)$
- $\text{eqs}(1) := \{(x \cdot e = x), (e \cdot x = x)\}$, $\text{eqs}(3) := \{x \cdot (y \cdot z) = (x \cdot y) \cdot z\}$

Model of a signature (sort, op, ar)

- a set X

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- for every operation $o \in \text{op}$ with a function $f_o \in X^{\text{ar}(o)} \rightarrow X$

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- a constant $e \in X$ and a function $f \in X \times X \rightarrow X$

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- for every operation $o \in \text{op}$ with a function $f_o \in X^{\text{ar}(o)} \rightarrow X$
→ lifted to $\llbracket _ \rrbracket \in \text{Tm}(n) \rightarrow X^n \rightarrow X$

A monoid = a model of the signature of monoids:

- a set X
- a constant $e \in X$ and a function $f \in X \times X \rightarrow X$
- $\llbracket e \rrbracket = e, \quad \llbracket (x \cdot e) \cdot y \rrbracket(u, v) = f(f(u, e), v)$

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- such that for every $(l, r) \in \text{eqs}(n)$ and $v \in X^n$, we have $\llbracket l \rrbracket(v) = \llbracket r \rrbracket(v)$

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- such that for all u, v and w we have $f(e, v) = v$, $f(v, e) = v$ and $f(u, f(v, w)) = f(f(u, v), w)$

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+ **Morphism of model**

Theorem (originally, Birkhoff?)

Every signature has a **free model**, which is initial amongst all models.

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Every signature has a **free model**, which is initial amongst all models.

Free model for the signature of monoids	=	free monoid = $\{e\}$
Free model for the signature of monoids + 1 constant per $a \in A$	=	free monoid on A

GENERALISED ALGEBRAIC THEORY

MULTI-SORTED UNIVERSAL ALGEBRA

Monoids, rings, propositional logic... fit in universal algebra

Modules, vector spaces... need multiple sorts

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Modules, vector spaces... need multiple sorts

Multi-sorted signature

A signature Σ consists of:

- a **set of sorts** $\text{sort}_{\Sigma} \in \text{Set}$
- a set of operations op_{Σ} , specified by an arity $\text{ar}_{\Sigma} \in \text{op}_{\Sigma} \rightarrow \text{sort}_{\Sigma}^* \times \text{sort}_{\Sigma} \rightarrow \text{Set}$
→ generates a set of terms $\text{Tm} \in \text{sort}_{\Sigma}^* \times \text{sort}_{\Sigma} \rightarrow \text{Set}$
- a set of equations $\text{eqs}_{\Sigma}(v, s) \subseteq \text{Tm}(v, s) \times \text{Tm}(v, s)$

Monoids, rings, propositional logic... fit in universal algebra

Modules, vector spaces... need multiple sorts

- two sorts = s for scalars and v for vectors
- operations $ar(\cdot_v) := ([s, v], v)$ $ar(\cdot_s) := ([s, s], s)$ $ar(+_s) := ([s, s], s)$ $ar(+_v) := ([v, v], v)$
- + equations

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→ generates a set of terms $Tm \in sort_\Sigma^* \times sort_\Sigma \rightarrow \mathbf{Set}$
- a set of equations $eqs_\Sigma(v, s) \subseteq Tm(v, s) \times Tm(v, s)$

STILL NOT ENOUGH!

But there's more!

- Category: set of morphisms $\text{Hom}(X, Y)$ for objects X, Y
- First-order logic: expressions, formulas over a given context
- Type theory: types over a context, terms of a type in a context

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- Category: set of morphisms $\text{Hom}(X, Y)$ for objects X, Y
- First-order logic: expressions, formulas over a given context
- Type theory: types over a context, terms of a type in a context

Missing ingredient = dependent sorts

A category:

- sorts $\text{ob} \in \mathbf{Set}$ and $\text{hom} \in \prod_{x,y \in \text{ob}} \mathbf{Set}$
- operations $\text{id} \in \prod_{x \in \text{ob}} \text{hom}(x, x)$ and $\circ \in \prod_{g \in \text{hom}(y,z), f \in \text{hom}(x,y)} \text{hom}(x, z)$
- equations $\prod_{f \in \text{hom}(x,y)} f \circ \text{id}(x) = f$ etc.

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- equations $\prod_{f \in \text{hom}(x,y)} f \circ \text{id}(x) = f$ etc.

A (model of) dependent type theory:

- $\text{Ctx} \in \mathbf{Set}$ (contexts)
- $\text{Ty} \in \text{Ctx} \rightarrow \mathbf{Set}$ (types)
- $\text{Tm} \in \prod_{\Gamma \in \text{Ctx}} \text{Ty}(\Gamma) \rightarrow \mathbf{Set}$ (terms)
- ...

Use a **type theory** (= the **logical framework**) to define signatures

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Signature, take 2

A signature is a context in the type theory below.

$$\frac{}{\Sigma \vdash \text{Sort}} \qquad \frac{\Sigma \vdash X : \text{Sort}}{\Sigma \vdash \text{El}(X)}$$
$$\frac{\Sigma \vdash A : \text{Sort} \quad \Sigma \vdash l : \text{El}(A) \quad \Sigma \vdash r : \text{El}(A)}{\Sigma \vdash l \equiv_A r} \quad (\text{extensional})$$

$$\Sigma, (X : \text{Sort})$$

$$\Sigma', (x : \text{El}(X))$$

$$\Sigma'', (e : x \equiv_A y)$$

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 \frac{\Sigma \vdash A : \text{Sort} \quad \Sigma, (x : A) \vdash B}{\vdash (x : A) \rightarrow B} \qquad \frac{\Sigma \vdash f : (x : A) \rightarrow B \quad \Sigma \vdash u : \text{El}(A)}{\Sigma \vdash f u : \text{El}(B[u])} \\
 \\
 \Sigma, (X : \text{Sort}) \qquad \Sigma', (f : \text{El}(X) \rightarrow \text{El}(X)) \qquad \Sigma'', (e : (x : \text{El}(X)) \rightarrow f(f x) \equiv_X x)
 \end{array}$$

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 \end{array}$$

For now: work with this type theory informally

THE GAT OF CATEGORIES, VERY FORMALLY

ob : Sort

hom : $(x: \text{ob}) \rightarrow (y: \text{ob}) \rightarrow \text{Sort}$

id : $(x: \text{ob}) \rightarrow \text{El}(\text{hom}(x, x))$

comp : $(x, y, z: \text{ob}) \rightarrow (f: \text{hom}(x, y)) \rightarrow (g: \text{hom}(y, z)) \rightarrow \text{El}(\text{hom}(x, z))$

idl : $(x, y: \text{ob}) \rightarrow (f: \text{hom}(x, y)) \rightarrow \text{comp}(x, x, y, \text{id}(x), f) \equiv_{\text{hom}(x, y)} f$

idr : $(x, y: \text{ob}) \rightarrow (f: \text{hom}(x, y)) \rightarrow \text{comp}(x, y, y, f, \text{id}(y)) \equiv_{\text{hom}(x, y)} f$

assoc : $(w, x, y, z: \text{ob}) \rightarrow (f: \text{hom}(w, x)) \rightarrow (g: \text{hom}(x, y)) \rightarrow (h: \text{hom}(y, z)) \rightarrow$
 $\text{comp}(x, y, w, f, \text{comp}(y, z, w, g, h)) \equiv_{\text{hom}(w, z)} \text{comp}(x, z, w, \text{comp}(w, x, y, f, g), h)$

THE GAT OF CATEGORIES, VERY FORMALLY

ob : Sort

hom : (x: ob) → (y: ob) → Sort

id : (x: ob) → El(hom(x, x))

comp : (x, y, z: ob) → (f: hom(x, y)) → (g: hom(y, z)) → El(hom(x, z))

idl : (x, y: ob) → (f: hom(x, y)) → comp(x, x, y, id(x), f) $\equiv_{\text{hom}(x,y)}$ f

idr : (x, y: ob) → (f: hom(x, y)) → comp(x, y, y, f, id(y)) $\equiv_{\text{hom}(x,y)}$ f

assoc : (w, x, y, z: ob) → (f: hom(w, x)) → (g: hom(x, y)) → (h: hom(y, z)) →
comp(x, y, w, f, comp(y, z, w, g, h)) $\equiv_{\text{hom}(w,z)}$ comp(x, z, w, comp(w, x, y, f, g), h)

What a mouthful!

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 $\circ : \{x, y, z:\text{ob}\} \rightarrow (g:\text{hom}(y, z)) \rightarrow (f:\text{hom}(x, y)) \rightarrow \text{El}(\text{hom}(x, z))$

$(x, y:\text{ob}) \rightarrow (f:\text{hom}(x, y)) \rightarrow f \circ \text{id}_x \equiv f$
 $(x, y:\text{ob}) \rightarrow (f:\text{hom}(x, y)) \rightarrow \text{id}_y \circ f \equiv f$
 $(w, x, y, z:\text{ob}) \rightarrow (f:\text{hom}(w, x)) \rightarrow (g:\text{hom}(x, y)) \rightarrow (h:\text{hom}(y, z)) \rightarrow$
 $(h \circ g) \circ f \equiv h \circ (g \circ f)$

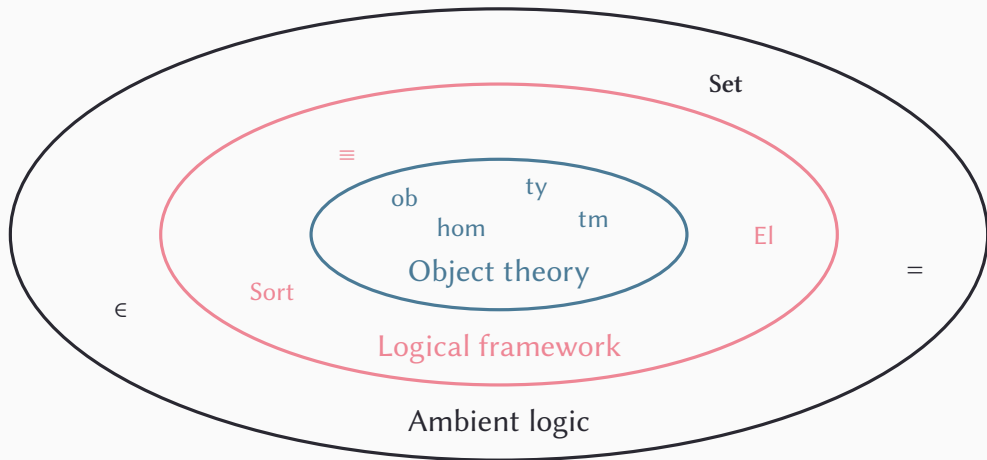
What a mouthful! Conventions: implicit arguments + notations

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 \frac{\vdash x : \text{ob}}{\vdash \text{id}_x : \text{hom}(x, y)} \qquad \frac{\vdash x, y, z : \text{ob} \quad \vdash f : \text{hom}(x, y) \quad \vdash g : \text{hom}(y, z)}{\vdash g \circ f : \text{hom}(x, z)} \\
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 \frac{\vdash x, y : \text{ob} \quad \vdash f : \text{hom}(x, y)}{\vdash f \circ \text{id}_x \equiv f : \text{hom}(x, y)} \qquad \frac{\vdash x, y : \text{ob} \quad \vdash f : \text{hom}(x, y)}{\vdash \text{id}_y \circ f \equiv f : \text{hom}(x, y)} \\
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 \frac{\vdash w, x, y, z : \text{ob} \quad \vdash f : \text{hom}(w, x) \quad \vdash g : \text{hom}(x, y) \quad \vdash h : \text{hom}(y, z)}{\vdash (h \circ g) \circ f \equiv h \circ (g \circ f) : \text{hom}(w, z)}
 \end{array}$$

What a mouthful! As typing rules

THE TYPE THEORIST'S FAVOURITE GAT: CATEGORIES WITH FAMILIES



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$\text{ctx} : \text{Sort}$

$\text{sub} : (\Gamma : \text{ctx}) \rightarrow (\Delta : \text{ctx}) \rightarrow \text{Sort}$

...

ctx and sub form a category

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ctx and sub form a category

$\text{ty} : (\Gamma : \text{ctx}) \rightarrow \text{Sort}$

$_[_] : \{\Gamma, \Delta : \text{ctx}\} \rightarrow (X : \text{ty}(\Delta)) \rightarrow (\sigma : \text{sub}(\Gamma, \Delta)) \rightarrow \text{El}(\text{ty}(\Gamma))$

$(\Gamma : \text{ctx}) \rightarrow (X : \text{ty}(\Gamma)) \rightarrow X[\text{id}] \equiv X$

...

ty is a functor

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$_[_] : \{\Gamma, \Delta : \text{ctx}\} \rightarrow \{X : \text{ty}(\Delta)\} \rightarrow (x : \text{tm}(\Delta, X)) \rightarrow (\sigma : \text{sub}(\Gamma, \Delta)) \rightarrow \text{El}(\text{tm}(\Gamma, X[\sigma]))$

$(\Gamma : \text{ctx}) \rightarrow (X : \text{ty}(\Gamma)) \rightarrow (x : \text{tm}(\Gamma, X)) \rightarrow x[\text{id}] \equiv_{\text{tm}(\Gamma, X)} x$

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tm is a displayed functor

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$\text{El}(\text{tr})$

$(\Gamma : \text{ctx})$

...

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functor

THE TYPE THEORIST'S FAVOURITE GAT: CATEGORIES WITH FAMILIES (CTD.)

$\diamond : \text{El}(\text{ctx})$

$! : \{\Gamma : \text{ctx}\} \rightarrow \text{El}(\text{sub}(\Gamma, \diamond))$

$(\Gamma : \text{ctx}) \rightarrow (\sigma : \text{sub}(\Gamma, \diamond)) \rightarrow \sigma \equiv !$

ctx has a terminal object (empty context)

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context extension $(\Gamma, x : X)$

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context extension $(\Gamma, x : X)$

$\text{var} : \{\Gamma : \text{ctx}\} \rightarrow \{X : \text{ty}(\Gamma)\} \rightarrow \text{El}(\text{tm}(\Gamma \triangleright X, X))$ last variable in context

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last variable in context

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weakening

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context extension $(\Gamma, x : X)$

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weakening

$(_, _) : \{\Gamma, \Delta : \text{ctx}\} \rightarrow \{X : \text{ty}(\Gamma)\} \rightarrow (\sigma : \text{sub}(\Delta, \Gamma)) \rightarrow$
 $(x : \text{tm}(\Delta, X[\sigma])) \rightarrow \text{El}(\text{sub}(\Delta, \Gamma \triangleright X))$

substitution extension

+ equations

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substitution extension

+ equations

Only terms: $\text{var}[\uparrow^n]$ = n -th variable in context

MODELS OF A GAT

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El(X)	$\mapsto$	$\llbracket X \rrbracket$
$\equiv$	$\mapsto$	$=$
$(_) \rightarrow _$	$\mapsto$	$\prod$
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A **model** of a GAT Σ is an inhabitant of $\llbracket \Sigma \rrbracket$.

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Definition

A **model** of a GAT Σ is an inhabitant of $\llbracket \Sigma \rrbracket$.

In fact, we have a **category** $\text{Mod}(\Sigma)$ (might be too strict to your liking!)

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Read from the GAT definition! Or...

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Definition (Fiore 2012, Awodey 2018)

A **natural model** consists of:

- a category $\mathcal{C}$ with a terminal object = contexts + substitutions + empty context
- two presheaves Ty and Tm (functors $\mathcal{C}^{op} \rightarrow \mathbf{Set}$) = types, terms, action of substitutions
- a natural transformation $ty : Tm \rightarrow Ty$ = maps a term to its type

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“ $\Gamma \triangleright A$ is the universal context which extends Γ and has a term of type A ”

$$\begin{array}{ccc} y(\Gamma \triangleright A) & \xrightarrow{\text{var}} & Tm \\ \uparrow \downarrow & \lrcorner & \downarrow ty \\ y\Gamma & \xrightarrow{A} & Ty \end{array}$$

WHAT IS A CwF?

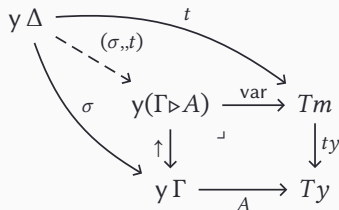
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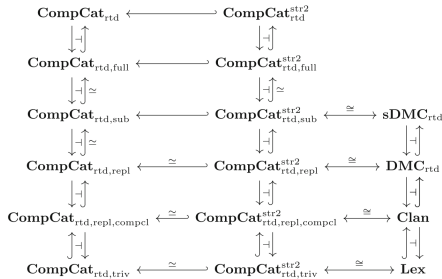


Fig. 4. Notions with types as maps, rooted

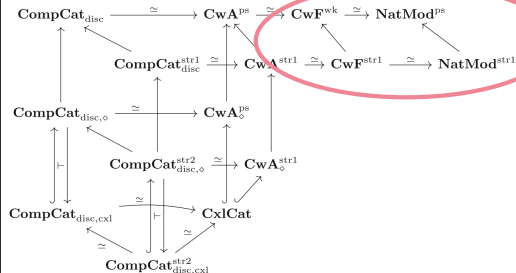


Fig. 5. Notions with types primitive

empty context
action of substitutions

SYNTAX

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Constructions that are valid in every model?

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Some sort of manipulation of syntax trees?

It is the same!

WHY ARE SYNTAX TREE NOT ENOUGH?

Ill-formed syntax trees: $0 + +$ $\lambda x.y$

Ill-typed syntax trees: $0 + (\lambda x.x)$

→ some syntax trees do not represent any valid terms

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Equations: $\text{id} \circ f = f$

→ many syntax trees represent equal terms

If f represents a valid term of type $\text{hom}(a, b)$, is $f \circ \text{id}_a$ valid?

→ equations play a role in typing!

Theorem (Cartmell 1986, Kaposi, Kovács and Altenkirch 2019)

The initial model of any GAT exists.

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Implementation: work with pre-terms, but check for typing/definitional equality

Much better if typing is **decidable**!

WRAPPING UP

One type theory to describe them all: a **logical framework**

Do the work **once**, get it for **all theories**:

- syntax
- (category of) models

Syntax is the **initial model**

WHERE TO GO FROM HERE?

To go on with GATs and CwFs

- *Generalised algebraic theories and contextual categories* (Cartmell 1986)
- *Categories with families: Unityped, simply typed, and dependently typed* (Castellan et al. 2021)
- *Constructing Quotient Inductive-Inductive Types* (Kaposi et al. 2019)
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Other logical frameworks

- SOGATs = GATs + built-in handling of contexts & variables
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Theorem (Uemura 2022)

The category of GATs is the bi-initial category with finite limits and an exponentiable morphism.

Thank you!

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